Experiment - 4.1.1. Set Operations

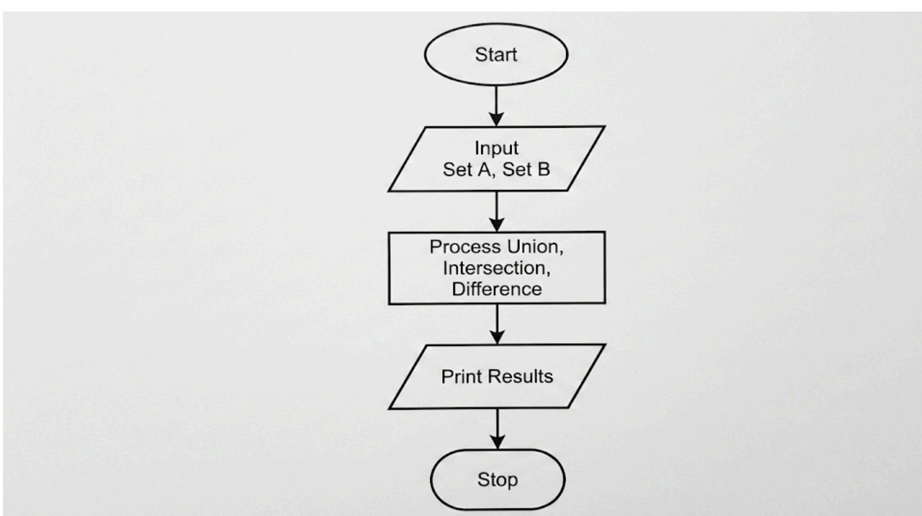
1. Aim

To design and implement a Python program that performs mathematical set operations, specifically union, intersection, and difference, on two sets: *Set A* and *Set B*. The program prompts the user to input space-separated integers for both sets and computes the results.

2. Pseudocode

1. **START**
2. **PRINT** the prompt "Set A: " and **READ** a line of space-separated integers.
3. **CONVERT** the input values into a mathematical set and **STORE** it in the variable A.
4. **PRINT** the prompt "Set B: " and **READ** a line of space-separated integers.
5. **CONVERT** the input values into a mathematical set and **STORE** it in the variable B.
6. **CALCULATE** the union of Set A and Set B (A | B) and **STORE** the result in union_set.
7. **CALCULATE** the intersection of Set A and Set B (A & B) and **STORE** the result in intersection_set.
8. **CALCULATE** the difference between Set A and Set B (A - B) and **STORE** the result in difference_set.
9. **PRINT** the text "Union:" followed by the value of union_set.
10. **PRINT** the text "Intersection:" followed by the value of intersection_set.
11. **PRINT** the text "Difference:" followed by the value of difference_set.
12. **END**

3. Flowchart



4. Python Program

```
# Read sets
A = set(map(int, input("Set A: ").split()))
B = set(map(int, input("Set B: ").split()))

# Operations
union_set = A | B
intersection_set = A & B
difference_set = A - B

# Output
print("Union:", union_set)
print("Intersection:", intersection_set)
print("Difference:", difference_set)
```

5. Experiment Screenshot